Cytiva
100 Results Way
Marlborough, MA 01752
United States

3 June, 2022

Re: AKTA ready Flow Kit Storage Conditions

To Whom It May Concern,

The recommended storage temperature for standard AKTA ready flow kits is provided in Section 8.3 of the Operating Instructions 28960345 and specified as > +5°C. This recommendation also applies to all modified AKTA ready flow kits as well, including the two listed in the below table. Extended storage below the recommended +5°C could lead to brittleness or cracking of the plastic connectors. However, the operating temperature of AKTA ready flow kits is +2°C to +40°C. If the kits are allowed to acclimate to a warmer temperature before being used this would reduce the risk of damage to the kit during setup and handling.

Description	**Part Number**	**Operating Temperature**
High Flow Kit F, Modified, AKTA ready	29477427	+2°C to +40°C
High Flow Gradient C, Modified, AKTA ready	29184612	+2°C to +40°C

Operating Instructions access link:
https://cdn.cytivalifesciences.com/api/public/content/digi-14501-pdf

This letter is provided for informational purposes only and does not affect or create any rights or obligations in regard to either party.

Sincerely,
Cytiva Quality Assurance

Certificate of Quality

This product is manufactured in compliance with our ISO 9001 certified quality management system.

Issued by Cytiva Westborough Quality Assurance
This document has been electronically produced and is valid without a signature.

Product:	AKTA ready Low Flow Kit
Lot Number:	18356721
Product Article Number:	28 9301 82
Date of Manufacture:	20240315
Product Description:	Low Flow Kit, AKTA ready
Expiration Date:	20260315

Product Release Criteria

We hereby certify that the defined product has been manufactured to meet its specifications and have been verified to meet predetermined Critical to Quality attributes.

Description	Test Method	Result
Autoclave - Pump tubing	121°C > 15 min	Conforms
Gamma Irradiation - Inlets	25.0 - 40.0 kGy	Conforms
Flow Rate Test	Per specification	Conforms
Pressure Integrity Test	Per specification	Conforms
Visual Inspection	Per specification	Conforms
Package Integrity	Per specification	Conforms

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Product:	AKTA ready High Flow Kit
Lot Number:	15102934
Product Article Number:	28 9301 83
Date of Manufacture:	20190712
Product Description:	High Flow Kit, AKTA ready
Expiration Date:	20210712

Product Release Criteria

We hereby certify that the defined product has been manufactured to meet its specifications and have been verified to meet predetermined Critical to Quality attributes.

Description	Test Method	Result
Autoclave - Pump tubing	121°C > 15 min	Conforms
Gamma Irradiation - Inlets	25.0 - 40.0 kGy	Conforms
Flow Rate Test	Per specification	Conforms
Pressure Integrity Test	Per specification	Conforms
Visual Inspection	Per specification	Conforms
Package Integrity	Per specification	Conforms

Packaging Component Specification

Document Number: PKG-SPEC-2023-0847
Revision: C
Effective Date: 2023-11-08

Component: AKTA ready Flow Kit Packaging Assembly
Drawing Reference: DWG-29477427-PKG-R3

1. Scope

This specification defines the packaging configuration for AKTA ready Flow Kit assemblies, including primary and secondary packaging components.

2. Materials

Component	Material	Specification
Blister Tray	PETG	ASTM D1003, min 92% clarity
Lid Film	Tyvek 1073B	Per DuPont specification
Secondary Carton	Corrugated	ECT-32, C-flute
Inner Cushion	PE Foam	2 lb density, 1 inch thick

3. Part Numbers

Assembly Part Number: PKG-29477427-C
Blister Tray: PKG-TRAY-4477-B
Lid Film: PKG-LID-1073-A
Secondary Carton: PKG-BOX-9301-A

Packaging Component Specification (continued)

4. Configuration Change History

Rev	Date	ECN Number	Change Description
A	2019-03-15	ECN-2019-0312	Initial release - PVC blister tray
B	2021-06-22	ECN-2021-0541	Updated tray insert dimensions per customer feedback (+3mm)
C	2023-11-08	ECN-2023-0847	Changed blister material from PVC to PETG (EU Reg 2023/1297)

5. Drawing Changes

Drawing DWG-29477427-PKG-R3 supersedes DWG-29477427-PKG-R2.
Key changes located on drawing:
- Sheet 1: Updated tray cavity dimensions (Section B-B)
- Sheet 2: New lid film sealing parameters (Detail C)
- Sheet 3: Updated label placement (per ECN-2023-0847)

Approved by: Packaging Engineering, Cytiva Uppsala
Date: 2023-11-08

Declaration Regarding Transmissible Spongiform Encephalopathies (BSE/TSE Compliance Statement)

Manufacturer: Cytiva Sweden AB
Address: Bjorkgatan 30, SE-751 84 Uppsala, Sweden

Product Family: AKTA ready Flow Kits
Applicable Part Numbers: 29477427, 29184612, 28930182, 28930183

Declaration

We hereby declare that the above-referenced products do not contain any materials of animal origin, including but not limited to bovine, ovine, caprine, or porcine derived materials.

All polymeric materials used in the manufacture of these products are of synthetic origin and comply with the following regulations:

- EU Regulation (EC) No 722/2012 (BSE/TSE risk assessment)
- EU Regulation (EC) No 1774/2002 (animal by-products)
- United States Pharmacopeia (USP) <87> and <88>

Risk Assessment Date: 15 March 2023
Assessment Valid Until: 15 March 2028
Assessment Reference: BSE-RA-2023-AKTA-001

This declaration is based on information provided by our raw material suppliers and our own assessment of the manufacturing process.

Signed: Quality Assurance Director
Cytiva Sweden AB
Date: 15 March 2023

Material Description Sheet

Product: AKTA ready Gradient Flow Section With Inlets
Part Number: 29184612
Revision: AD

1. Product Description

The AKTA ready Gradient Flow Section is a pre-assembled flow path component designed for use with AKTA chromatography systems. The flow section includes inlet manifolds, pump tubing, and gradient mixing chambers.

2. Materials of Construction

Component	Material	Specification
Housing	Polypropylene (PP)	ISO 1873-1, FDA 21 CFR 177.1520
Gaskets	EPDM Rubber	USP Class VI, FDA 21 CFR 177.2600
Pump Tubing	Platinum-Cured Silicone	USP Class VI, FDA 21 CFR 177.2600
Inlet Fittings	PEEK	ASTM D6262
Mixing Chamber	Borosilicate Glass	USP Type I, ASTM E438

3. Sterilization Compatibility

Method	Conditions	Compatible
Autoclave	121°C, 15 min	Yes (tubing)
Gamma Irradiation	25-40 kGy	Yes (fittings)
EtO Sterilization	Not validated	No

4. Physical Properties

Dimensions: 245 x 120 x 85 mm
Weight: 380 g (assembled)
Operating Temperature: +2°C to +40°C
Operating Pressure: Max 5 MPa
Shelf Life: 2 years from date of manufacture

Supplier Qualification Record

Supplier Name: Cytiva Sweden AB
Supplier Address: Bjorkgatan 30, SE-751 84 Uppsala, Sweden
Supplier Code: SUP-2019-0847
Qualification Status: Approved

1. Qualification History

Initial Qualification Date: 14 September 2019
Last On-Site Audit: 20 November 2023
Audit Result: Approved (no critical findings)
Next Scheduled Audit: November 2025

2. Certifications

Certification	Number	Valid Until
ISO 9001:2015	QMS-SE-2021-4847	2024-12-31
ISO 13485:2016	MD-SE-2021-1293	2024-12-31
FDA Establishment	3007488211	Active
EU Authorized Rep	AR-2020-CYTIVA	Active

3. Approved Products

Part Number	Description	Category
29477427	High Flow Kit F, Modified, AKTA ready	Flow Kit
29184612	High Flow Gradient C, Modified	Flow Kit
28930182	Low Flow Kit, AKTA ready	Flow Kit
28930183	High Flow Kit, AKTA ready	Flow Kit
29021085	AKTA ready Gradient Low Flow Section	Flow Section
29021086	AKTA ready Gradient High Flow Section	Flow Section

Supplier Qualification Record (continued)

4. Supply Chain Information

Manufacturing Site: Cytiva Eysins Facility, Eysins, Switzerland
Secondary Site: Cytiva Uppsala Facility, Uppsala, Sweden
Warehouse: Cytiva Distribution Center, Marlborough, MA, USA

Sub-Supplier Information:

Sub-Supplier	Material Provided	Qualification Status
DuPont (Tyvek)	Lid film (1073B)	Approved (2020)
Wacker Chemie AG	Silicone tubing	Approved (2019)
Sabic Innovative	Polypropylene resin	Approved (2021)
Schott AG	Borosilicate glass	Approved (2019)

5. Quality Agreement

Quality Agreement Reference: QA-2019-CYTIVA-0847
Agreement Date: 14 September 2019
Last Reviewed: 20 November 2023
Key Terms: Annual audit rights, CAPA notification within 48 hours, change notification 90 days prior to implementation.

6. Performance Metrics (Last 12 Months)

On-Time Delivery: 97.3%
Incoming Quality: 99.8% (2 minor NCRs, 0 critical)
CAPA Response Time: Average 12 days

Cytiva
100 Results Way
Marlborough, MA 01752
United States

Global Chain of Custody

List of Assemblies Manufactured at Cytiva Eysins Facility in Switzerland

Cytiva Item #	Description
28930182	Low Flow Kit, AKTA ready
28930183	High Flow Kit, AKTA ready
29021085	AKTA ready Gradient Low Flow Section
29021086	AKTA ready Gradient High Flow Section
29477427	High Flow Kit F, Modified, AKTA ready
29184612	High Flow Gradient C, Modified, AKTA ready

All listed assemblies follow the same chain of custody:
1. Raw materials received at Eysins facility (incoming QC)
2. Sub-assembly and final assembly (in-process checks)
3. Final quality release (per product-specific release criteria)
4. Shipped to Cytiva Distribution Center, Marlborough, MA
5. Distributed to end customer upon order

Traceability: Each assembly carries a unique lot number traceable to raw material lots, manufacturing date, and operator records.

Document prepared by: Supply Chain Quality, Cytiva
Date: 15 January 2024